OMM Lab #27

Diagnosing the Hip, Tibiofemoral, Fibula, Tibiotalar, Talocalcaneal, Tarsal, and Tarsometatarsal Joints

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The Femoroacetabular Joint (Hip)

The major motions of the hip (femoroacetabular) joint that the patient can voluntarily create and are grossly tested are shown below along with ranges and approximate means for a 30- to 40-year-old population:

- **Flexion** 120 degrees (range: 90 to 150 degrees)–**extension** 10 degrees (range 0 to 35 degrees)
- **Abduction** 40 degrees (range: 15 to 55 degrees)–**adduction** 30 degrees (15 to 45 degrees)
- **External rotation** 35 degrees (range: 10 to 55 degrees)–**internal rotation** 30 degrees (range: 20 to 50 degrees)

Minor joint motions of the hip assessed with end-feel palpation include

- Anterior glide occurring with external rotation
- Posterior glide occurring with internal rotation

Gross hip ranges of motion and passive assessments of end feel are usually tested with the patient supine and the pelvis stabilized. Ranges of motion vary from individual to individual and differ in different populations tested. For example, expect different averages when testing male collegiate football players compared to teenage female gymnasts. Therefore, a side-to-side comparison of both symmetry and the quality of the barrier at the end of motion is often much more important than measuring the absolute number of degrees in the range of motion. This holds true for the assessment of all lower extremity joints. In the presence of asymmetric motion, the quality and pattern of the barriers help to differentiate hypomobility on one side from hypermobility on the other hand to distinguish somatic dysfunction from pathology.

The character and pattern of the end feel of each hip motion may be springy (physiologic) or it might suggest a capsular, traumatic, or dysfunctional pattern. Subtle differences in somatic dysfunction of myofascial rather than articular origin may also be appreciated by evaluating the quality of the end feel of the hip joint.

It is critical to be able to discern between loss of range of motion due to somatic dysfunction, which is amenable to OMT, or to a primary hip pathology that requires medical or surgical care. The physical exam can help to clarify this distinction.

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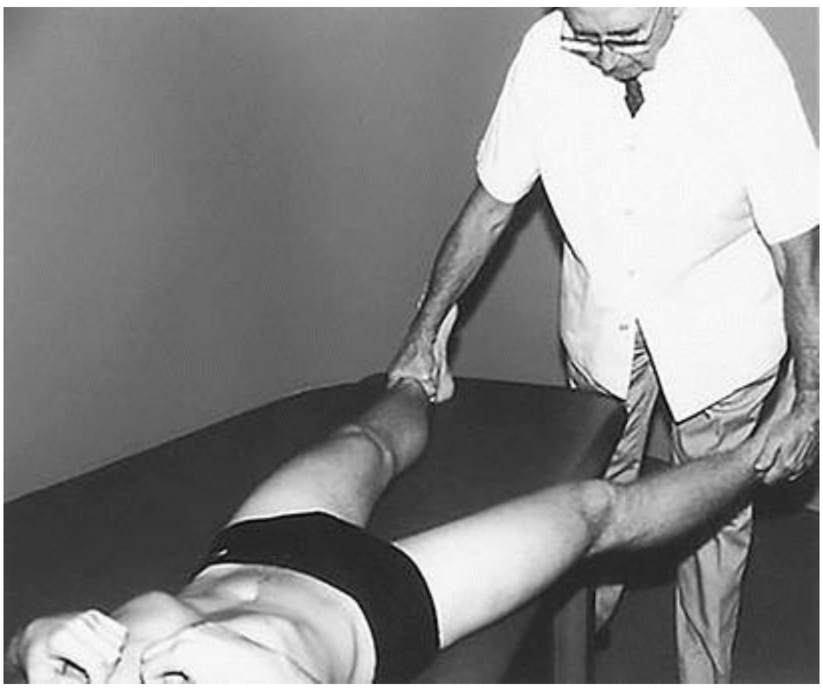
Testing the hip adductors
(Abduction ROM)

1. Patient lies supine on the examination table.
2. Physician stands at the end of the table.
3. Physician grasps the leg to be examined at the ankle, lifts it slightly, and abducts the leg, noting the range and quality of the motion.
4. Steps 1 to 3 are repeated with the patient's opposite leg.
5. Physician compares the degree of abduction of the involved leg with that of the opposite leg.



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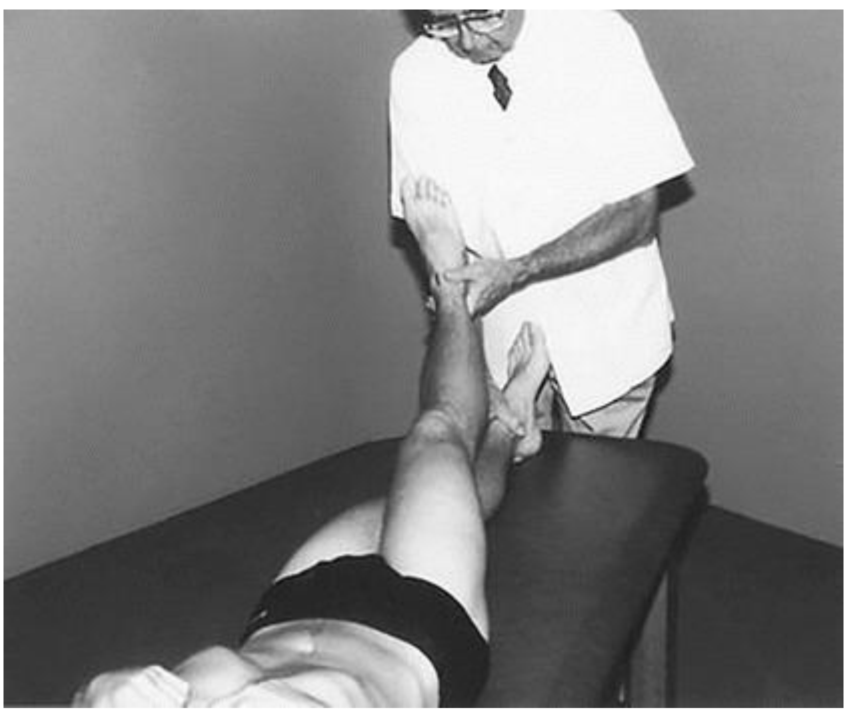
Testing the hip abductors

(Adduction ROM)

1. Patient lies supine on the examination table.
2. Physician stands at the end of the table.
3. Physician grasps the leg to be examined at the ankle, lifts it slightly, and adducts it in front of the opposite leg, noting the range and quality of the motion.
4. Steps 1 to 3 are repeated with the patient's opposite leg.
5. Physician compares the degree of adduction of the involved leg with that of the opposite leg.



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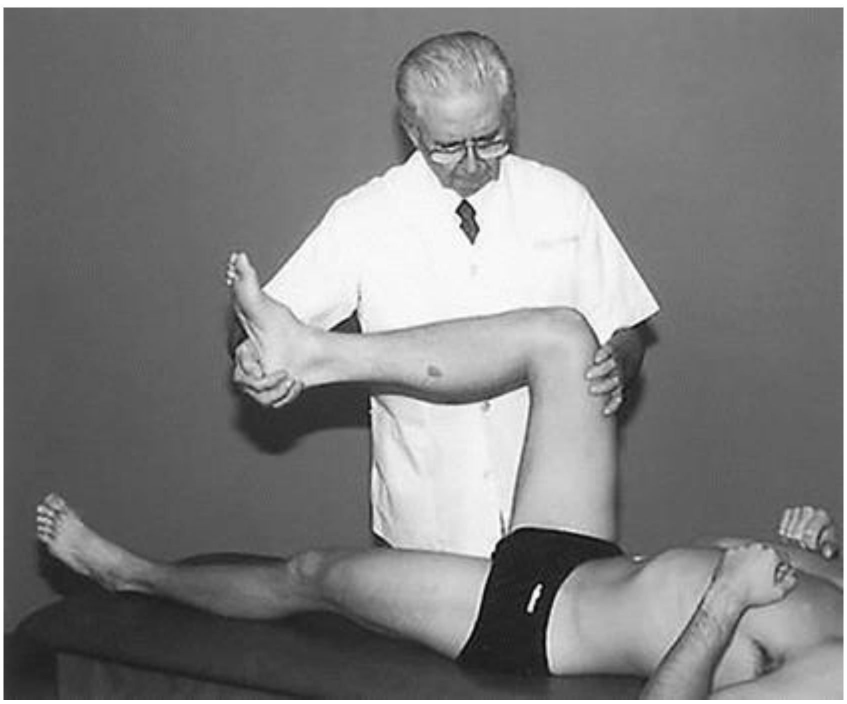
Testing the hip internal rotators (External Rotation ROM)

1. Patient lies supine on the examination table.
2. Physician stands on the side of the extremity to be tested.
3. Physician grasps the leg to be tested and flexes both the hip and the knee, each to 90 degrees.
4. Physician then externally rotates the femur by moving the foot and ankle medially, noting the range and quality of motion.
5. Steps 1 to 4 are repeated with the patient's opposite leg.
6. Physician compares the degree of external rotation of the involved leg with that of the opposite leg.



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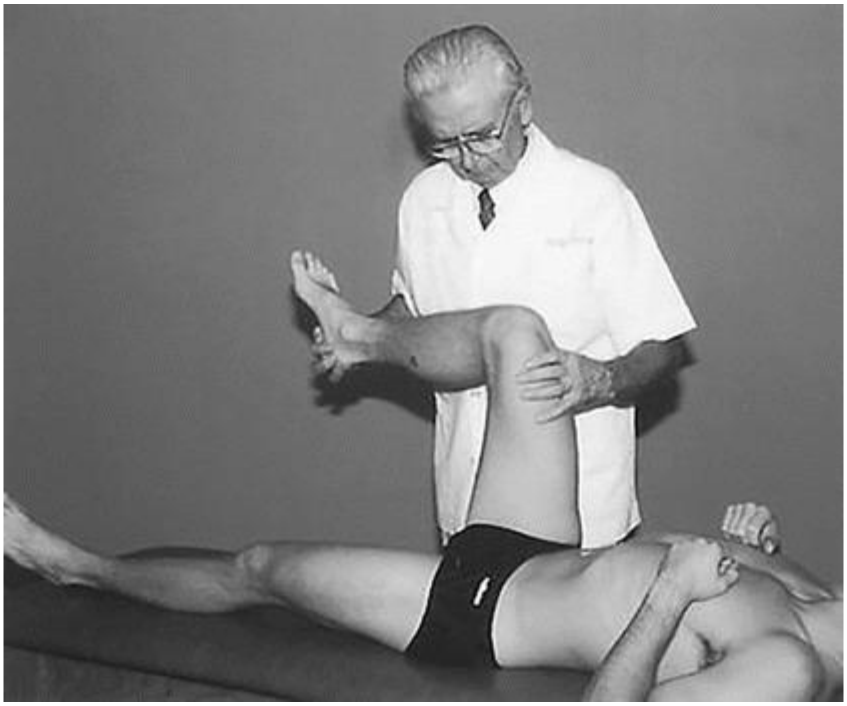
Testing the hip external rotators (Internal Rotation ROM)

1. Patient lies supine on the examination table.
2. Physician stands on the side of the extremity to be tested.
3. Physician grasps the leg to be tested and flexes both the hip and the knee, each to 90 degrees.
4. Physician then internally rotates the femur by moving the foot and ankle laterally, noting the range and quality of motion.
5. Steps 1 to 4 are repeated with the patient's opposite leg.
6. Physician compares the degree of internal rotation of the involved leg with that of the opposite leg.



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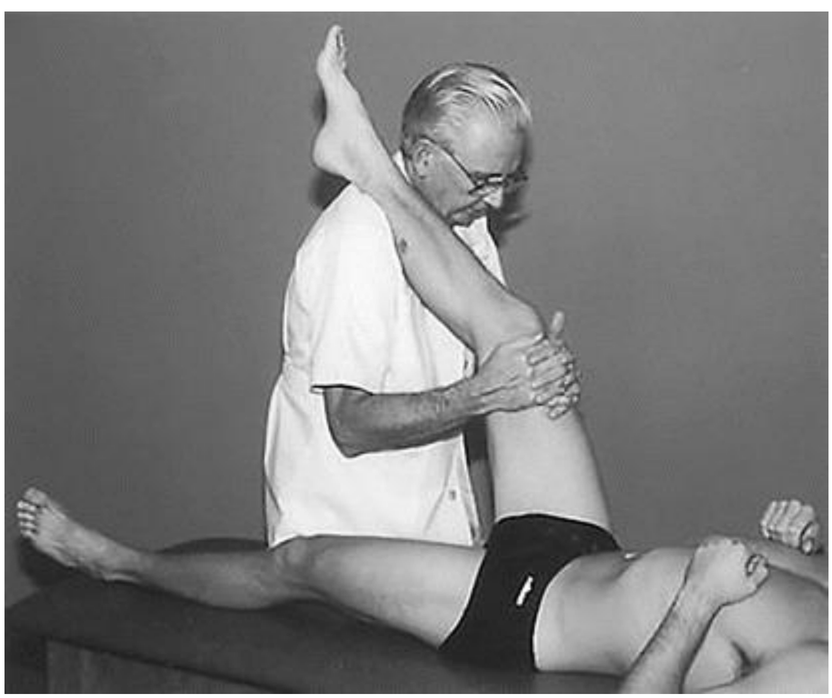
Testing the hip extensors (Flexion ROM)

1. Patient lies supine on the examination table.
2. Physician stands on the side of the extremity to be tested.
3. Physician grasps the ankle of the leg to be tested with one hand.
4. The pads of the physician's fingers, on the physician's other hand, are placed on the anterior superior iliac spine on the patient's opposite side.
5. Physician lifts the involved leg, introducing hip flexion, until motion is noted at the opposite anterior superior iliac spine. The range and quality of the motion is noted.
6. Steps 1 to 5 are repeated with the patient's opposite leg.
7. Physician compares the degree of hip flexion of the involved leg with that of the opposite leg.



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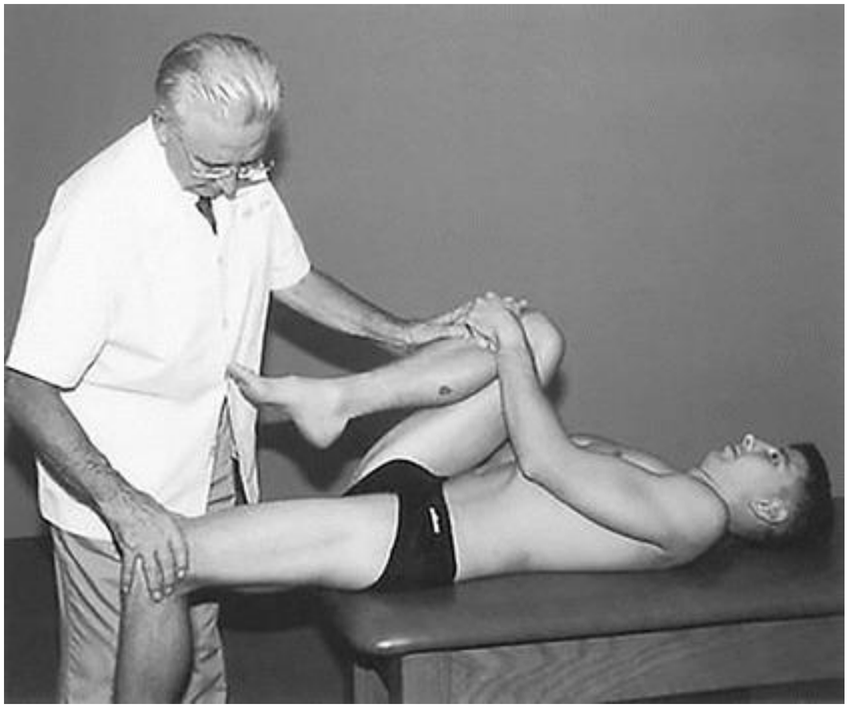
Testing the hip flexors (iliopsoas muscles) (Extension ROM aka Thomas Test)

1. Patient lies supine on the examination table with the pelvis at the end of the table, so that the lower extremities below the knees can hang over the edge of the table.
2. Physician stands at the end of the examination table, facing the patient.
3. Patient flexes both hips and knees, bringing the knees close to the chest.
4. Patient grasps one knee and holds the leg close to the chest, while the physician passively extends the leg to be tested.
5. The range and quality of the leg motion is noted. Normal motion is for the thigh to rest on the table surface with the knee fully flexed.
6. Steps 1 to 5 are repeated with the opposite leg.

7. Physician compares the degree of hip extension of the involved leg with that of the opposite leg



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Tibiofemoral Joint (knee)

- Flexion SD with anterior glide
- Extension SD with posterior glide
- Internal rotation SD with posterolateral glide
- External rotation SD with anteromedial glide

Complete extension of the knee creates a bony lock. Testing of minor motions of the joint and ligament assessment should therefore be performed with the knee in variable degrees of flexion. Remember SD is most often found in the minor motions/glides.

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Extension (NYITCOM OMM Department Media)

Kimberly manual 2nd Edition(2008) page 291

Checking Internal and External rotation of tibia ((NYITCOM OMM Department Media))

Fibular Motion

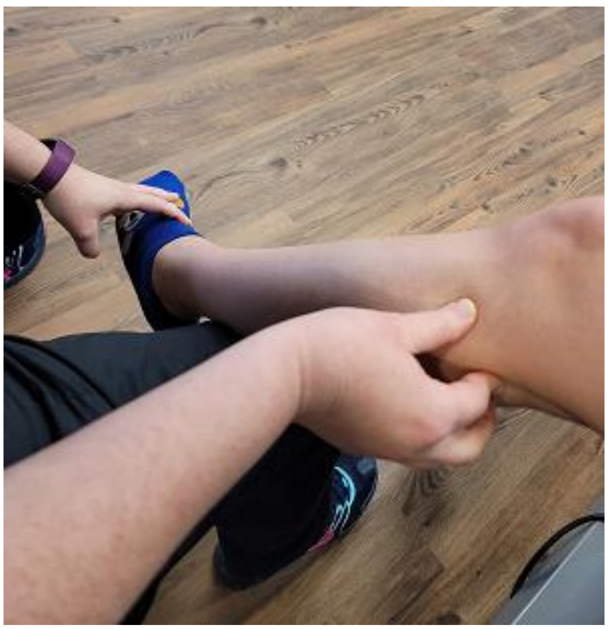
- Anterior Proximal Fibula SD = anterolateral glide
- Posterior Proximal Fibula SD = posteromedial glide

When the fibular head glides anteriorly, reciprocal motion is initiated at the distal fibula (lateral malleolus), which glides posteriorly. Posterior fibular head motion is accompanied by anterior motion at the distal fibula.

External rotation of the tibia and ankle carries the distal fibula posteriorly and elevates and glides the proximal fibular head anteriorly. The opposite occurs with internal rotation of the tibia and ankle.

With pronation of the foot/dorsiflexion of the talus, ligamentous attachments glide the distal fibula posteriorly with reciprocal glide of the proximal head of the fibula anteriorly. The opposite occurs with supination of the foot/plantar flexion of the talus.

Plantar flexion = posterior fibular head



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Dorsiflexion = anterior fibular head

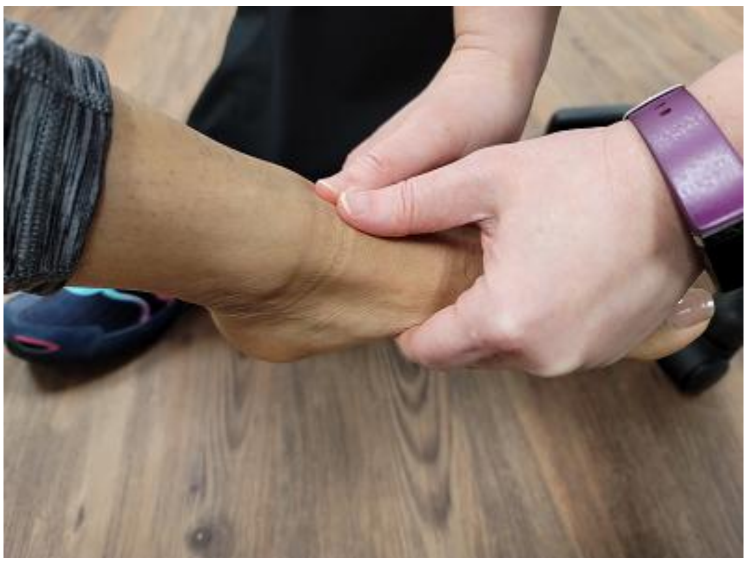
Tibiotalar Joint (talus)

- Plantar flexion with anterior glide
- Dorsiflexion with posterior glide

Adduction, toeing-in, and some supination of the foot accompany plantar flexion. This motion carries the lateral malleolus anteriorly. Through reciprocal action of the fibula, the proximal fibular head also glides posteriorly and inferiorly. The talus glides anteriorly, placing the narrow portion of the talus in the ankle mortise, a less stable position.

Abduction, toeing-out, and some pronation of the foot accompany dorsiflexion. This type of motion carries the lateral malleolus posteriorly and, through reciprocal action, glides the fibular head anteriorly and superiorly. With dorsiflexion, the talus glides posteriorly. Because the talus is structurally wider anteriorly, it fits more securely with the posterior glide component in the ankle mortise. Dorsiflexion is, because of its structure, a more functionally stable position. This stability is the reason taping techniques to treat or prevent ankle sprains usually emphasize a dorsiflexed position.

Plantar flexion



Dorsiflexion

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Talocalcaneal Joint (Subtalar)

- Inversion SD
- Eversion SD

Movement of the calcaneus produces leg rotation:

Inversion of the calcaneus produces external rotation of the tibia and causes the talus to glide posterolaterally over the calcaneus. Eversion of the calcaneus produces internal rotation of the tibia, and anteromedial glide of the talus on the calcaneus.

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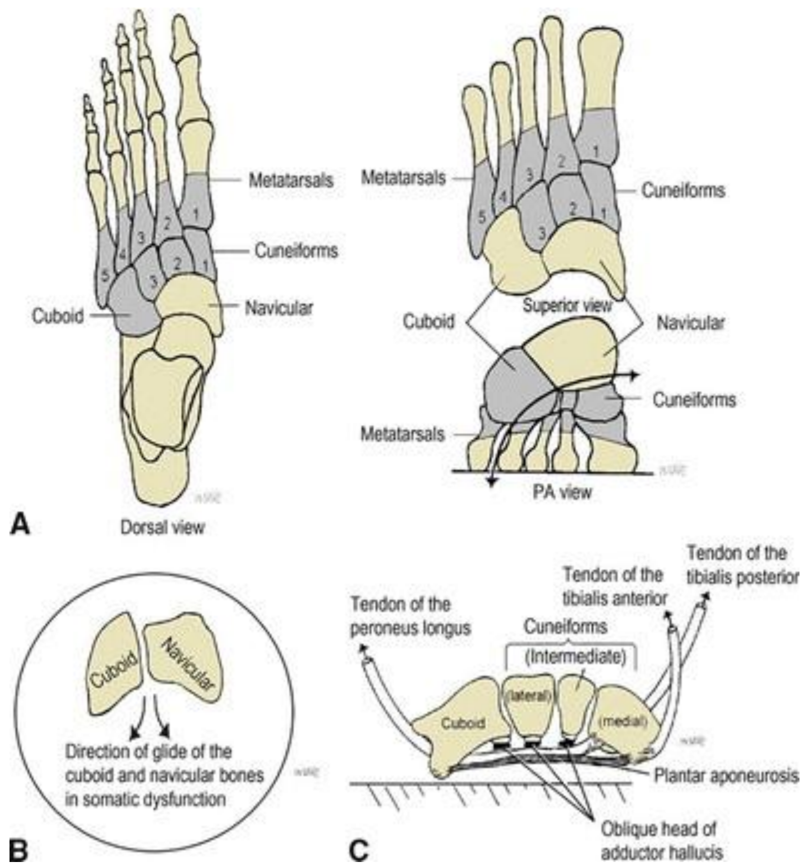
1. With one hand stabilize the talus
2. Cup the patient's heel in the palm of the other hand
3. Introduce inversion and eversion of the calcaneus
4. Name SD for preference of motion: Either inversion or eversion

Kimberly Manual 2nd Edition (2008) page 305

Tarsal Somatic Dysfunction



- Plantar Glide SD
- Dorsal Glide SD

Somatic dysfunction of the cuboid involves the edge nearest the middle of the foot. In SD this edge most commonly glides toward the plantar surface of the foot and rotates laterally around its AP axis; somatic dysfunction of the navicular most commonly involves the edge nearest the middle of the foot gliding toward the plantar surface and rotating medially around its AP axis. Cuneiform somatic dysfunction most commonly manifests as a consequence of the second cuneiform gliding directly plantar ward. Somatic dysfunction of these tarsal bones can be diagnosed by the combination of tenderness and increased tissue tension over the plantar and/or dorsal surface of each of these bones.

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Cuneiforms

Cuboid and Navicular

Tarsometatarsal Joints

Somatic dysfunction of the tarsometatarsal, metatarsophalangeal, and interphalangeal joints involves their minor motions:

- Plantar or dorsal glide
- Internal or external rotation
- Lateral or medial glide

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1. With one hand stabilize the tarsal bones.
2. With the other hand grasp the metatarsal you want to diagnose and check for the preferred motion: plantar/dorsal, internal/external rotation, and lateral/medial glide
3. Name SD for preferred motions. Should have 3.

Kimberly Manual 2nd Edition (2008) page 308

Review Material: all PPOM-1 Innominates/Pelvic Region Lab Material – diagnosis and any treatment.